Experimental Study of a Stabilized 230-GHz Gyrotron-Driver for Frequency Locking of Megawatt-Level Gyrotrons

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***Abstract*— This brief presents the results of experimental study of a 230-GHz gyrotron-driver designed for the frequency locking of a prototype megawatt-power gyrotron for plasma heating and current drive in the next generation of fusion power plants, such as tokamaks TRT and DEMO. The driver provides the output power of up to 130 kW at $TE_{18,8}$ operating mode in a free-running regime. The frequency tuning dependencies on the magnetic field, cavity temperature, accelerating, and anode voltage are measured. When the generation frequency is stabilized using a phase-locked loop system controlling a modulating anode voltage, the radiation power is reduced to 60 kW. The obtained power and frequency tuning are sufficient for a fairly wide (∼40–50 MHz) locking band of the implemented megawatt gyrotron.**

***Index Terms*— Frequency locking, gyrotrons, high-power microwaves, stabilization.**

I. Introduction

As IS known, gyrotrons are the most in-demand sources of powerful microwave radiation for electron-cyclotron plasma heating in modern installations aimed at achieving controlled thermonuclear fusion (CTF) [1], [2], [3], [4], [5]. The general trend in the development of CTF systems is to increase the frequency and power of each gyrotron, as well as the total heating power due to an increase in the number of radiation sources [3], [5]. In order to maintain an acceptable ohmic load with an increase in the gyrotron power, it is necessary to expand the transverse size of the cavity. Together with frequency rise, this leads to an extremely dense spectrum in the operating frequency region. This results in a strong mode competition, which complicates the gyrotrons operating conditions, and, in some cases, does not allow reaching the calculated parameters of output power and efficiency. In addition, the suppression of plasma instabilities requires phase synchronization of several gyrotrons [6].

According to [7], [8], [9], and [10], the indicated problems can be solved by phase and frequency locking (PFL) of a megawatt gyrotron with a relatively weak (several percent of power) external signal. In this case, the excitation of parasitic modes is suppressed, which allows achieving the optimal range of parameters for high-power operation. The development of an internal two-directional quasi-optical converter [11] for inputting an external signal into a gyrotron made it possible to experimentally demonstrate the PFL operation of a 170 GHz/1 MW gyrotron [12], [13]. An increase in power and efficiency, as well as stabilization of the frequency of the slave MW tube were observed. For these experiments, a specialized input signal source (gyrotron-driver [14]) with a power level of up to 25 kW and a stabilized frequency based on a phased-locked loop (PLL) system was developed.

At present, taking into account the abovementioned trend of increasing the frequency of electron-cyclotron heating in the DEMO [4] and TRT [15] installations, a 230-GHz megawatt-power gyrotron with the $TE_{33,13}$ operating mode has been implemented at the IAP RAS [16]. Based on previous experimental data with the 170-GHz gyrotron and numerical simulation, we believe that the output power and efficiency of the 230-GHz tube can be significantly increased in the PFL regime. This brief presents the results of experimental tests of a new stabilized gyrotron-driver with operating frequency of 230 GHz and increased (compared to [14]) output power, intended for frequency locking of the specified above higher-frequency megawatt-power gyrotron.

II. Concept of a 230-GHz Gyrotron-Driver With a Phase-Locked Loop

According to theoretical analysis [7], [8], [9], [10], to ensure effective frequency locking, the power of the gyrotron-driver can be 2%–5% of the power of the slave megawatt-level gyrotron. However, increasing the driver power up to 10%

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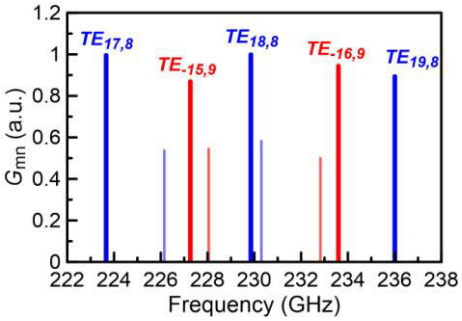

Fig. 1. Coupling coefficients $G_{m,p}$ of competing waveguide $TE_{m,p}$ modes (mode spectrum) in the cyclotron resonance band near the operating frequency of 230 GHz.

or more is useful for expanding the locking band, as well as (when divided into several channels) for synchronizing several tubes at once. At the same time, the maximum size of the electron-optic and electrodynamic systems is limited by the available magnetic system, namely the cryogen-free magnet JMTD-10T100 with a hot bore diameter of 100 mm and a magnetic field strength of up to 10 T. With a sufficiently high expected output power of up to 100 kW and higher, the $TE_{18,8}$ mode was chosen as the operating one, for which, with a radius of the regular part of the resonator $R_0 = 9.88$ mm, acceptable ohmic loads on the copper walls are ensured. The spectrum of competing modes in the cyclotron resonance band with the coupling coefficient values $G_{m,p}$ for the optimal electron beam injection radius $R_b = 3.965$ mm is shown in Fig. 1. As can be seen, a typical situation is realized, when the most dangerous competitors are co- and counter-rotating modes of the equidistant spectra. According to theoretical analysis, a sufficiently large intermode interval allows us to expect single-mode single-frequency generation with excitation of the operating mode at an accelerating voltage U of 47–50 kV and a beam current I_b of up to 20 A.

It should be noted that, as in [14], the key design feature of the described gyrotron-driver is the triode configuration of the magnetron-injection gun, in which the modulating anode is an isolated electrode with a relatively small electrical capacitance of 100 pF. Accordingly, the gyrotron operation can be quickly controlled due to a rather small change in the anode voltage U_a, which, nevertheless, leads to a significant variation of the radiation frequency due to a strong change in the susceptibility of the electron beam [17], [18]. The frequency stabilization of the gyrotron is provided on the basis of the PLL system, the scheme of which is described in detail in [19] and was also used for the 170-GHz gyrotron-driver [14]. In this scheme, the collector and resonator are grounded; a high negative voltage of up to −50 kV is applied to the cathode and a rather small (0–2 kV) negative voltage controlled by a feedback loop [20] is applied to the anode. The control signal is generated by a phase detector, which compares the signal from a harmonic mixer with a signal from a reference oscillator.

III. Results of Experimental Tests

At the first stage of the experiments, the gyrotron was tested in the free-running regime, that is, without the PLL control.

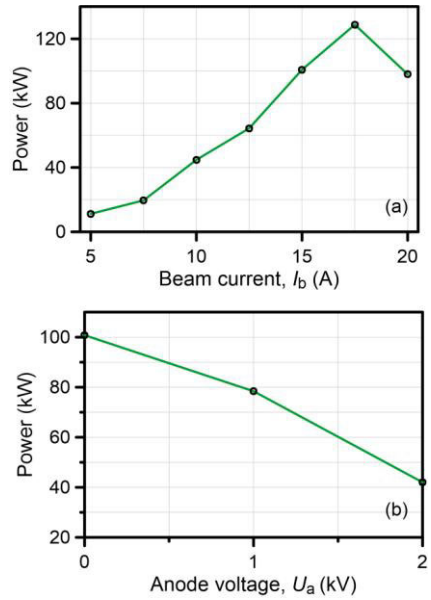

Fig. 2. Results of experimental tests of a 230-GHz gyrotron-driver without PLL frequency stabilization. Dependences of the maximum radiation power P (a) on the beam current I_b at accelerating voltage $U_0 = -48$ kV and (b) on the anode voltage U_a at $U = -48$ kV and $I_b = 15$ A.

Fig. 2(a) shows the dependences of the radiation power on the electron beam current I_b at accelerating voltages $U = -48$ kV for the optimal values of the guiding magnetic field. The maximum measured power was about 130 kW at 17.5 A. For the parameters $I_b = 15$ A and $U = -48$ kV, corresponding to the output power of 100 kW, the influence of the anode voltage U_a on the generation power was analyzed. According to the theory of formation of rotating electron beams for gyrotrons, applying control voltage to the anode leads to a decrease in the pitch factor of the beam, which is the ratio of the rotational and translational velocities of electrons. As a result, there is a decrease in the output power [Fig. 2(b)].

Since the possibility of adjusting the frequency of the external signal is of great importance in the process of locking a high-power slave gyrotron, the dependencies of the gyrotron-driver generation frequency on the anode voltage, as well as on the temperature t of the resonator cooling water were measured. The frequency of the radiation for the gyrotron-driver was measured using the 17th harmonic of the heterodyne frequency $f_{LO} = 13563$ MHz provided by a Keysight E8257D generators, the intermediate frequency f_{IF} signal was measured using Keysight DSAZ594A oscilloscope: $f_{IF} = 17 f_{LO} - f_{driver}$. The results are shown in Fig. 3. As can be seen from [Fig. 3(a)], the frequency adjustment by temperature is about 5 MHz/°C, or about 100 MHz when the resonator temperature changes from 15 to 35 °C. Fig. 3(b) shows the generation frequency for different anode voltages. It can be seen that the frequency tuning is on the order of 80 MHz per 1 kV. Performed frequency measurements demonstrate the possibility of perfect matching of the gyrotron-driver with a recently implemented 1-MW gyrotron, for which the frequency in the maximum efficiency region is 230.214 GHz.

At the next stage of the experiments, the operation of the gyrotron-driver was studied in the frequency stabilization

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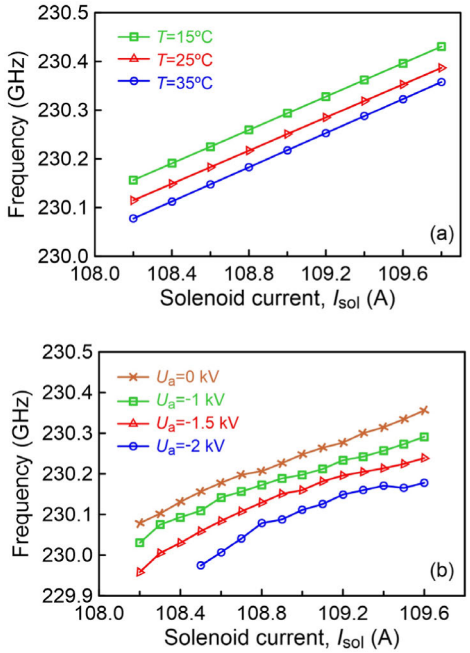

Fig. 3. Radiation frequency versus solenoid current I_{sol} (guiding magnetic field) (a) for different temperatures of the cooling water ($U = -48$ kV, $I_b = 10$ A, $U_a = 0$ kV) and (b) for different anode voltages ($U = -48$ kV, $I_b = 10$ A, $t = 25$ °C).

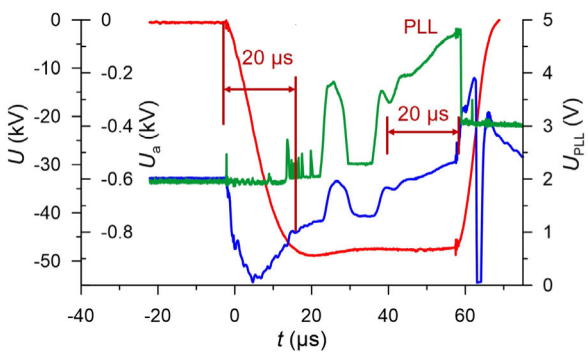

Fig. 4. Oscillograms of the accelerating voltage U pulse (red), the PLL control signal U_{PLL} (green), and the anode voltage U_a (blue).

regime provided by PLL control of modulating anode voltage. Fig. 4 demonstrates the oscillograms of the accelerating voltage pulse (red) with the amplitude of −48 kV, the PLL control signal (green), and the anode voltage (blue).

In fact, it is necessary to understand that, frequency stabilization is possible within a certain band, defined by the frequency of the reference oscillator, gyrotron frequency sensitivity to anode voltage and parameters of PLL. In the experiment, the PLL locking bandwidth was 5 MHz, and locking time was about 10 μs. During the accelerating voltage rise (first 20 μs of the pulse), the radiation frequency passes the locking band of the PLL, which corresponds to the first peak of the control signal U_{PLL}. When the voltage U reaches its steady-state value (approximately at the 30th microsecond), the gyrotron frequency returns to the locking band and the PLL locks for the rest of the pulse duration. Thus, the total duration of the stabilized operation is about 20 μs.

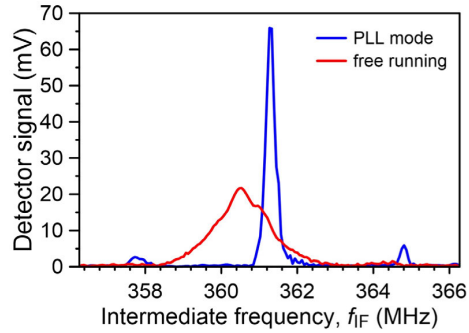

Fig. 5. Comparison of the measured output radiation spectra in free-running operation (red) of the gyrotron-driver and in the PLL stabilized regime (blue).

TABLE I
RADIATION FREQUENCY [GHz]/OUTPUT POWER [kW] OF THE PLL-STABILIZED GYROTRON DRIVER AT DIFFERENT VALUES OF THE SOLENOID CURRENT (GUIDING MAGNETIC FIELD) AND COOLING WATER TEMPERATURES ($U = -48$ kV, $I_B = 15$ A)

	I_{sol}, A			
T, °C	**108.4**	**107.6**	**108.8**	**109.0**
15	230.192/42	230.214/36		
25	230.161/44		230.214/30	
35	230.129/42			230.214/14

The output radiation spectra in various operating regimes are shown in Fig. 5. In the PLL regime, the measured spectrum width is determined by the duration of stabilized operation; thus, the specified width cannot be less than 0.4 MHz. Nevertheless, a sharp narrowing of the main spectral line was observed compared to the free-running generation with spectrum width of about 2 MHz.

By simultaneously adjusting the temperature of the resonator and the magnetic field, the radiation frequency and output power of the gyrotron-driver can be tuned in a wide range. Table I presents both the tuning of the output frequency at constant power and variation of output power at a constant frequency. Such capabilities would allow to investigate the dependencies of the locking zone width of the megawatt gyrotron on the power and frequency of the locking signal and verify the existing theories, describing the operation of a frequency-locked gyrotron.

IV. CONCLUSION

Thus, we have implemented and experimentally tested the gyrotron-driver with the operating frequency of 230 GHz, intended for frequency-locking of a megawatt-power gyrotron prototype for plasma heating in the TRT and DEMO tokamaks. The PLL stabilization of the generation frequency in short pulses has been demonstrated with the wide tuning range of both frequency and output power. The output power in the PLL regime is about 60 kW, which is sufficient to lock the frequency of the megawatt gyrotron recently developed at IAP RAS.

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